Kiyan Tavangar

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Research Interests:

I study dark matter using observations and simulations of stars in the Milky Way and the Local Group. I have used stellar streams and phase-space spirals to understand galactic dynamics in the Milky Way halo and disk, respectively. I am interested in continuing that work while also applying the tools and intuition I have gained to other dynamical questions in an effort to constrain dark matter.

Education

PhD Candidate in Astronomy at Columbia University **Sep 2022 – May 2027 (expected)**
Thesis Title: *Constraining dark matter substructure using stellar substructures in the Milky Way: streams in the halo and snails in the disk*
Advisors: Kathryn V. Johnston, Adrian Price-Whelan
Awards: National Osterbrock Leadership Program Fellow (2023-2025), Lead Teaching Fellow (2024-2025)

Provost Scholar at the University of Chicago, Chicago, IL **Graduated: 06/21**
Astrophysics B.S. (Honors, Cum Laude), Mathematics B.S.

Teaching and Outreach

Co-instructor for *Physics for Poets* Spring 2026
I co-taught *Physics for Poets* with Professor Jeremy Dodd at Taconic State Correctional Facility. This is a Columbia course that incarcerated students are taking for credit. Professor Dodd and I split the work for the course approximately equally.

Instructor of *Introduction to Research Skills in Astrophysics* Spring 2024
I developed and am teaching a course to help undergraduates interested in astronomy get started with research. The course is designed for undergraduates with no research experience. This course is heavily influenced by the success of the StreamTeamTNG program described below.

Organizer and Instructor for the STEM Prison Outreach Program Feb 2024 – present
I founded a STEM outreach program at MDC Brooklyn and continue to be the lead organizer. We have biweekly events led by a rotating cast of Columbia scientists who present an overview of their field. In addition to recruiting volunteers, I also generally lead the first session of each semester with an introduction to astronomy.

Co-organizer of Communiverse in the Columbia University Astronomy Department Sep 2024 – present
Communiverse is a biweekly event in the astronomy department that focuses on teaching, pedagogy, and science communication. As a co-organizer, I help create the schedule and occasionally lead discussions myself.

Lead Teaching Fellow at Columbia University Aug 2024 – June 2025
The Lead Teaching Fellows design and run pedagogical workshops twice a year, among other responsibilities. I ran workshops on how to assign work in the age of AI and on mentoring in STEM.

Teaching Assistant for undergraduate *Stars and Atoms* course Spring 2023
Course for non-majors of all years. I hosted office hours weekly and helped design the final project.

Teaching Assistant for undergraduate *Galaxies* course Fall 2022
Course for Astronomy or Astrophysics majors (usually third years). I hosted office hours weekly

Mentoring

Jessie Wang, *Columbia University,* (undergraduate) May 2025 – Dec 2025
Project: Exploring the feasibility of using stellar stream velocity dispersion profiles to identify stream progenitors
Co-advisor: Kathryn Johnston

Joshua Kalish, *Columbia University,* (undergraduate) May 2025 – present

Project: Exploring the evolution of mean dynamical properties for different regions of a simulated disk in response to a satellite perturbation

Co-advisors: Axel Widmark and Kathryn Johnston

StreamTeamTNG program, *Columbia University and CUNY,* (undergraduates) Jan 2023 – Aug 2023

Program: StreamTeamTNG was an extracurricular program designed to teach undergraduates interested in astronomy how to do research. It used example projects in stellar streams to teach the basics of working with simulations and data, reading papers, and making presentations. I designed most assignments and led some group meetings.

Publications

ORCID: 0000-0001-6584-6144

First Author or Co-first Author Publications

1. **Tavangar, K.,** et al. (in prep.). *Thalweg: An Updated Flexible Empirical Stream Modeling Framework Applied to ATLAS-Aliqa Uma With Gaia and DES.*

2. **Tavangar, K.,** Johnston, K. V., Hunt, J. A. S., Widmark, A., Kaushik, V. G., & Price-Whelan, A. M. (2026). Phase-Spirals Across Galactic Disks II: Using large-scale "macro-spirals" in phase-spiral amplitude to derive perturbation times. *The Astrophysical Journal, 1010(2),* 126, IOP. https://doi.org/10.3847/1538-4357/aea6b1

3. **Tavangar, K**., Johnston, K. V., Hunt, J. A. S., Widmark, A., Hamilton, C., Petersen, M. S., & Weinberg, M. D. (2026). Phase Spirals across Galactic Disks. I. Exploring Dynamical Influences on Winding. *The Astrophysical Journal*, *1005*(1), 120, IOP. https://doi.org/10.3847/1538-4357/ae7b28

4. Chiti, A., **Tavangar, K.**, Ferguson, P. S., Carballo-Bello, J. A., Senkevich, A. M., Erkal, D., et al. (2025). DELVE-ing into the Milky Way's Globular Clusters: Assessing Extratidal Features in NGC 5897, NGC 7492, and Testing Detectability with Deeper Photometry. *The Astronomical Journal*, *170*(5), 294, IOP. https://doi.org/10.3847/1538-3881/ae07e0

5. **Tavangar, K.**, & Price-Whelan, A. M. (2025). Inferring the Density and Membership of Stellar Streams with Flexible Models: The GD-1 Stream in Gaia Data Release 3. *The Astrophysical Journal*, *988*(1), 45, IOP. https://doi.org/10.3847/1538-4357/addd1c

6. **Tavangar, K.**, Ferguson, P., Shipp, N., Drlica-Wagner, A., Koposov, S., Erkal, D., et al. (2022). From the Fire: A Deeper Look at the Phoenix Stream. *The Astrophysical Journal*, *925*(2), 118, IOP. https://doi.org/10.3847/1538-4357/ac399b

Co-authored Publications with Meaningful Contributions

7. Frankel, N., Semczuk, M., Antoja, T., Chakrabarti, S., Chiba, R., Grand, R. J. J., et al. (2026). Winding, Unwinding, Rewinding the Gaia Phase Spiral. *Publications of the Astronomical Society of the Pacific*, *138*(6), 066001, IOP. https://doi.org/10.1088/1538-3873/ae6bee
   Contribution: I attended the conference which this paper summarizes and wrote the section about the phase spiral mentoring program, which I am leading, along with a few other minor additions.

8. Sanderson, R. E., McKinnon, K. A., Thob, A. C. R., Williams, B., **Tavangar, K**., Pace, A. B., et al. (2026). *A near field guide to Roman's wide-area surveys*. *arXiv e-prints* (arXiv:2603.12981). https://doi.org/10.48550/arXiv.2603.12981
   Contribution: I wrote the section on GD-1 and created Figure 9

9. Widmark, A., **Tavangar, K.**, Kalish, J., Johnston, K. V., & Hunt, J. A. S. (2026). The Phase Spiral's Origin and Evolution: Indications from Its Varying Properties across the Milky Way Disk. *The Astrophysical Journal*, *999*(1), 52, IOP. https://doi.org/10.3847/1538-4357/ae38dd
   Contribution: I had weekly discussions with the lead author about direction of project and I created Figure 11

10. Starkman, N., Nibauer, J., Bovy, J., Webb, J. J., **Tavangar, K.**, Price-Whelan, A., & Bonaca, A. (2025). Stream Members Only: Data-driven Characterization of Stellar Streams with Mixture Density Networks. *The Astrophysical Journal*, *980*(2), 253, IOP. https://doi.org/10.3847/1538-4357/ad94f2
    Contribution: I was leading an analysis with similar objectives but using a different method. The lead authors and I discussed the merits of each method and I provided suggestions for the analysis and the paper.

11. Ferguson, P. S., Shipp, N., Drlica-Wagner, A., Li, T. S., Cerny, W., **Tavangar, K.**, et al. (2022). DELVE-ing into the Jet: A Thin Stellar Stream on a Retrograde Orbit at 30 kpc. *The Astronomical Journal*, *163*(1), 18, IOP. https://doi.org/10.3847/1538-3881/ac3492
    Contribution: I helped create the morphological model used to characterize Jet in this paper.

12. Khullar, G., Gozman, K., Lin, J. J., Martinez, M. N., Matthews Acuña, O. S., Medina, E., et al. (2021). COOL-LAMPS. I. An Extraordinarily Bright Lensed Galaxy at Redshift 5.04. *The Astrophysical Journal*, *906*(2), 107, IOP. https://doi.org/10.3847/1538-4357/abcb86
    Contribution: I searched for strongly lensed objects in the DECaLS data, helped with the source modeling. participated in most discussions for this paper. Note: Authors 3-12 (including **Tavangar, K.)** contributed equally and are listed in alphabetical order

13. Shipp, N., Price-Whelan, A. M., **Tavangar, K.**, Mateu, C., & Drlica-Wagner, A. (2020). Discovery of Extended Tidal Tails around the Globular Cluster Palomar 13. *The Astronomical Journal*, *160*(5), 244, IOP. https://doi.org/10.3847/1538-3881/abbd3a
    Contribution: I quantitatively verified stream significance to support its discovery and created Figure 1 showing the discovered tidal tails

Co-authored Publications with Minor Contributions

14. Chiti, A., Drlica-Wagner, A., Pace, A. B., Cerny, W., Atzberger, K. R., Barbosa, F. O., et al. (2026). *The DECam MAGIC Survey $-$ Mapping the Ancient Galaxy in CaHK: Overview and Summary of Early Science*. *arXiv e-prints* (arXiv:2605.26581). https://doi.org/10.48550/arXiv.2605.26581
    Contribution: I am a MAGIC collaboration member and contributed to discussions when planning the survey

15. Cloonan, A. P., Khullar, G., Napier, K. A., Gladders, M. D., Dahle, H., Rosener, R., et al. (2025). COOL-LAMPS. VIII. Known Wide-separation Lensed Quasars and Their Host Galaxies Reveal a Lack of Evolution in $M_{BH}/M_{\cdot}$ Since z ~ 3. *The Astrophysical Journal*, *987*(2), 194, IOP. https://doi.org/10.3847/1538-4357/addabf
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

16. Mork, S. D., Gladders, M. D., Khullar, G., Sharon, K., Chicoine, N., Cloonan, A. P., et al. (2025). COOL-LAMPS. VII. Quantifying Strong-lens Scaling Relations with 177 Cluster-scale Strong Gravitational Lenses in DECaLS. *The Astrophysical Journal*, *979*(2), 184, IOP. https://doi.org/10.3847/1538-4357/ada24c
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

17. Klein, M., Sharon, K., Napier, K., Gladders, M. D., Khullar, G., Bayliss, M., et al. (2024). COOL-LAMPS. VI. Lens Model and New Constraints on the Properties of COOL J1241+2219, a Bright z = 5 Lyman Break Galaxy and its z = 1 Cluster Lens. *The Astrophysical Journal*, *963*(1), 44, IOP. https://doi.org/10.3847/1538-4357/ad22de
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

18. Aganze, C., Pearson, S., Starkenburg, T., Contardo, G., Johnston, K. V., **Tavangar, K.**, et al. (2024). Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope. *The Astrophysical Journal*, *962*(2), 151, IOP. https://doi.org/10.3847/1538-4357/ad159c
    Contribution: I discussed the analysis and results with the lead author and provided suggestions about how to connect this extra-galactic analysis with what has been done in the Milky Way.

19. Zhang, Y., Manwadkar, V., Gladders, M. D., Khullar, G., Dahle, H., Napier, K. A., et al. (2023). COOL-LAMPS. IV. A Sample of Bright Strongly Lensed Galaxies at 3 < z < 4. *The Astrophysical Journal*, *950*(1), 58, IOP. https://doi.org/10.3847/1538-4357/acc9be
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

20. Martinez, M. N., Napier, K. A., Cloonan, A. P., Sukay, E., Gozman, K., Merz, K., et al. (2023). COOL-LAMPS. III. Discovery of a 25.″9 Separation Quasar Lensed by a Merging Galaxy Cluster. *The Astrophysical Journal*, *946*(2), 63, IOP. https://doi.org/10.3847/1538-4357/acbe39
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

21. Sukay, E., Khullar, G., Gladders, M. D., Sharon, K., Mahler, G., Napier, K., et al. (2022). COOL-LAMPS. II. Characterizing the Size and Star Formation History of a Bright Strongly Lensed Early-type Galaxy at Redshift 1.02. *The Astrophysical Journal*, *940*(1), 42, IOP. https://doi.org/10.3847/1538-4357/ac9974
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

22. Drlica-Wagner, A., Ferguson, P. S., Adamów, M., Aguena, M., Allam, S., Andrade-Oliveira, F., et al. (2022). The DECam Local Volume Exploration Survey Data Release 2. *The Astrophysical Journal Supplement Series*, *261*(2), 38, IOP. https://doi.org/10.3847/1538-4365/ac78eb
    Contribution: I did some observing for the DELVE dataset and am a collaboration member

23. Li, T. S., Ji, A. P., Pace, A. B., Erkal, D., Koposov, S. E., Shipp, N., et al. (2022). S^5: The Orbital and Chemical Properties of One Dozen Stellar Streams. *The Astrophysical Journal*, *928*(1), 30, IOP. https://doi.org/10.3847/1538-4357/ac46d3
    Contribution: I contributed to the S^5 target selection and determining the stream features of some analyzed streams

24. Drlica-Wagner, A., Carlin, J. L., Nidever, D. L., Ferguson, P. S., Kuropatkin, N., Adamów, M., et al. (2021). The DECam Local Volume Exploration Survey: Overview and First Data Release. *The Astrophysical Journal Supplement Series*, *256*(1), 2, IOP. https://doi.org/10.3847/1538-4365/ac079d
    Contribution: I did some observing for the DELVE dataset and am a collaboration member

25. Stringer, K. M., Drlica-Wagner, A., Macri, L., Martínez-Vázquez, C. E., Vivas, A. K., Ferguson, P., et al. (2021). Identifying RR Lyrae Variable Stars in Six Years of the Dark Energy Survey. *The Astrophysical Journal*, *911*(2), 109, IOP. https://doi.org/10.3847/1538-4357/abe873
    Contribution: I searched for main sequence stellar populations associated with RRL clumps

26. Mau, S., Cerny, W., Pace, A. B., Choi, Y., Drlica-Wagner, A., Santana-Silva, L., et al. (2020). Two Ultra-faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey. *The Astrophysical Journal*, *890*(2), 136, IOP. https://doi.org/10.3847/1538-4357/ab6c67
    Contribution: I did some observing for the dataset and worked on the satellite search tools used here

Presentations

Invited Talks

1. Yale Galaxy Lunch — *Streams and Snails: Galactic Dynamics in the Halo and Disk* — Oct 2026
2. DiRAC Lunch Seminar — *Streams and Snails: Galactic Dynamics in the Halo and Disk* — Sep 2026
3. Lunch Seminar at Carnegie Observatories — *Streams and Snails: Galactic Dynamics in the Halo and Disk* — Sep 2026
4. Streams24: The Theory Edition — *Galstreams and CATS Tutorial* (with Cecilia Mateu) — Aug 2024
5. Columbia chapter of the Society of Physics Students — *Galactic Cannibalism: Milky Way Formation and Hints Towards Dark Matter.* — Oct 2024

Contributed Talks

6. How stars move around and what they tell us about galaxies' masses formation and evolution - From the Milky Way to giant elliptical galaxies — *Timing the Phase Spirals' Origin Using Large-Scale Correlations in Their Morphologies.* (poster) July 2026
7. Galactic Bars, Bulges and Disks — *Exploring dynamical influences on phase spiral winding across galactic disks in simulations* June 2026
8. Hernquist Group meeting at the CfA — *Using Phase Spirals to Understand the Milky Way – Sgr Merger* May 2026
9. Winding, Unwinding and Rewinding the Gaia Phase Spiral — *Correlating phase spirals and macro-snails using BFEs and mSSA* Aug 2025
10. Gravity in the Local Group — *Inferring the density and membership of stellar streams: Application to GD-1* June 2025
11. AAS245 — *Macro and Micro Snails from a Single Satellite Flyby with EXP* Jan 2025
12. Waves in the Milky Way Disk — *Macro and Micro Snails from a Single Satellite Flyby with EXP* Sep 2024
13. Survey Science Group meeting at the UChicago — *Modeling the Spur in the GD-1 Stream: A way to test the dark matter sub halo population.* Apr 2022
14. Streams21: Constraints on Dark Matter — *Stream Modeling of ATLAS and Phoenix in DES* Jan 2021

Outreach Talks (invited)

15. Westchester Amateur Astronomers — *Stellar Streams as Galactic Cannibals* Apr 2027
16. Columbia Astronomy Public Lectures — *Stellar Streams as Galactic Cannibals* Nov 2026

Scientific Collaborations

EXP Collaboration Student Member 2022 – present

Community Atlas of Tidal Streams (CATS) 2022 – present

Center for Computational Astrophysics (CCA) Dynamics Group at the Flatiron Institute 2021 – 2022

ChicagO Optically selected strong Lenses – Located At the Margins of Public Surveys Collaboration (COOL-LAMPS) Member 2020 – present

Southern Stellar Stream Spectroscopic Survey (S^5) Member 2020 – present

Dark Energy Survey (DES) Collaboration External Collaborator 2019 – present

DECam Local Volume Exploration Survey (DELVE) Student Member 2019 – present

Kavli Institute of Cosmological Physics Survey Science Group Member 2019 – 2021

Technical Proficiencies

Experience writing open-source software
Extensive knowledge of Python
Working knowledge of MATLAB, Stan, and Scheme
Experience in lensed source modeling
Extensive experience in dynamical, empirical, and probabilistic modeling
Extensive experience using large datasets
Remote observing experience with the Blanco 4m Telescope for the DELVE survey